Band-1 abc Quality Bound Probe (Collapse 2.24)

Christopher Baird + ZoraASI Band-1 dash (executor)

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Band-1 abc Quality / Restricted Radical Range Probe (Finite + Lit)

Honesty stamp. The **abc conjecture remains OPEN** in the sense of community consensus. This note does **not** claim abc proved. It does **not** treat Mochizuki's Inter-universal Teichmüller theory (IUT) as an accepted proof. The mathematical community has **not** accepted IUT as settling abc (Scholze–Stix objections; 2026 Project LANA interim report still suspends judgment). **No Clay / no ToE. GATE-04** locked. The box result is a **finite search**, **not** a quality supremum theorem and **not** a radical-range collapse proof.

abc is OPEN (community). Finite high-quality hits $\neq$ proof. **IUT $\neq$ accepted proof.**

Abstract

Literature (statement): For coprime positive integers $a + b = c$, the quality is

$$q(a, b, c) = \frac{\log c}{\log \operatorname{rad}(abc)},$$

where $\operatorname{rad}(n)$ is the product of the distinct prime factors of n. One equivalent form of the Oesterlé–Masser **abc conjecture** asserts that for every $\varepsilon > 0$ there are only finitely many such triples with $q(a, b, c) > 1 + \varepsilon$.

Literature (record quality): The standing computational record high quality remains Eric Reyssat's triple

$$2 + 3^{10} \cdot 109 = 23^5, \qquad q \approx 1.6299$$

(still cited as the maximum in 2026 secondary literature). Broader catalogs (ABC@Home lineage; Bart de Smit) list millions of $q > 1$ hits with c up to 10^{18} and beyond; **no accepted analytic proof** that a universal finite max exists outside the conjecture.

Literature (IUT): Mochizuki's IUT papers (claimed 2012; published PRIMS 2021) are **not** community-accepted as a proof of abc. Scholze–Stix identify a severe gap; Project LANA (2026 interim) reports no consensus / suspended judgment. **This probe does not claim abc via IUT.**

Box finite check: Exhaustive enumeration of coprime $a + b = c$ with $3 \le c \le 20{,}000$. Best on-box quality $q \approx 1.567887$ at $(1, 4374, 4375)$ (classical de Weger example). Label: **finite search only**.

Problem (collapse Band-2 item 2.24 → Band-1-style ledger)

1. State abc / quality $q(a, b, c)$ cleanly from the literature.
2. Ledger known record high-quality triples and computational search lineage.
3. State IUT status honestly (**not accepted**).
4. Certify a feasible on-box finite quality search; report best q in-window.
5. Keep Clay, ToE, GATE-04, and "IUT proves abc" off the claim surface.

Literature ledger (web check; not re-proved on box)

Year / item	Result	Status
1985–88	Oesterlé–Masser formulate abc	classical conjecture
quality form	$q = \log c / \log \mathrm{rad}(abc)$; finitely many $q > 1 + \varepsilon$	equivalent formulation
Reyssat	$2 + 3^{10} \cdot 109 = 23^5$, $q \approx 1.6299$	**standing record quality**
de Weger	e.g. $1 + 2 \cdot 3^7 = 5^4 \cdot 7$, $q \approx 1.5679$	classic high-quality
Stewart–Yu et al.	unconditional $c < \exp(K \cdot \mathrm{rad}^{\theta})$ bounds	proven, far from abc
ABC@Home / de Smit	millions of $q > 1$ triples (c to 10^{18}+)	computational catalogs
2012–2021	Mochizuki IUT claim; PRIMS publication	**not community-accepted proof**
2018	Scholze–Stix gap report	mainstream objection
2026	Project LANA interim: judgment suspended; no consensus	**abc still OPEN (community)**

Hygiene. Do **not** elevate IUT, Clay-millennial framing, or ToE language. Computational catalogs do not prove the finiteness-for-every-ε statement.

Method (box)

1. Precompute $\mathrm{rad}(n)$ for all $n \le C_{\max} = 20{,}000$ via linear sieve.
2. For each $c \in [3, C_{\max}]$ and $a \in [1, \lfloor c/2 \rfloor]$ with $\gcd(a, c-a) = 1$, set $b = c - a$.
3. Pairwise coprimality $\Rightarrow \mathrm{rad}(abc) = \mathrm{rad}(a)\mathrm{rad}(b)\mathrm{rad}(c)$.
4. Compute $q = \log c / \log \mathrm{rad}(abc)$; track best and all $q \ge 1.2$.
5. Spot-check known literature triples (including Reyssat outside window).
6. Label: **finite search** — not a bound on $\sup q$, not abc.

Results

Literature status

Quantity	Value
Conjecture (community)	**OPEN**
IUT as accepted proof	**No**
Record published/cited quality	≈ 1.6299 **(Reyssat)**
Unconditional analytic abc	**not available** (only weak exp bounds)

Box window (tonight)

Quantity	Value
$C_{\max}$	20,000
Coprime triples checked	60,795,197
Hits with $q > 1$	204

Quantity	Value
Best box quality	≈ 1.567887
Best triple	(1, 4374, 4375) with rad = 210
Wall (box)	≈ 20.3 s
Label	finite search $\neq$ proof; $\neq$ world catalog depth

Top on-box qualities ($q \geq 1.2$, selected)

q (approx)	(a, b, c)	rad(abc)
1.567887	$(1, 4374, 4375)$	210
1.455673	$(1, 2400, 2401)$	210
1.426565	$(3, 125, 128)$	30
1.360723	$(625, 2048, 2673)$	330
1.337552	$(289, 6272, 6561)$	714

Reyssat ($q \approx 1.6299$) lies **outside** this $c \leq 20{,}000$ window ($c = 23^5 = 6{,}436{,}343$); spot-check on box recovers $q \approx 1.62991$.

Why Band-1 vs conjecture

- **Band-1 (this probe):** lit ledger on quality + computational records; honest IUT status; certify finite $C_{\max}$ search.
- **Not claimed:** proof of abc; existence of a universal max q; IUT correctness; Clay/ToE; GATE-04 topics.
- **Restricted radical / quality bound angle:** the conjecture predicts that qualities cannot stay above any fixed $1 + \varepsilon$ infinitely often; this probe only **samples** small-c behavior and cites the known record — it does **not** establish a collapsed radical-range theorem.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/abc_quality/`
- Script: `abc_quality_search.py` → `results.json`
- Ledger: `literature_ledger.json`
- Rerun: `python3 abc_quality_search.py`

Conclusion (one sentence)

Finite box best $q \approx 1.5679$ (de Weger in-window); literature record $q \approx 1.6299$ (Reyssat); abc remains OPEN and IUT is not accepted as a proof.